SUNFLASER
OPTOELECTRONIC SOLUTIONS

Product Manual

Product Instruction
Parameter Description

SFA1000C

905 nm Miniature Laser Ranging Module

Disclaimer

- This product manual mainly provides the reference specifications and performance parameters of the product. If any specifications are not included in this document, please contact SunFlaser Optics Technology (Chengdu) Co., Ltd. directly for confirmation. Any changes to the contents of this manual or the specific specifications of the ranging module will not be notified separately.
- Without prior written permission from SunFlaser Optics Technology (Chengdu) Co., Ltd., this document shall not be reproduced, copied, or stored in any information storage system. It shall not be used for any unauthorized purposes or forwarded to any third party without explicit authorization from SunFlaser Optics Technology (Chengdu) Co., Ltd.
- Due to variations in actual product performance, this document is provided as an important reference for product specifications and parameters, but does not constitute any form of warranty. Any express or implied warranties regarding the product, including suitability for a specific purpose, shall be subject to the actual performance of the specific product.
- SunFlaser Optics Technology (Chengdu) Co., Ltd. reserves the right to revise this document without prior notice. This document may contain inaccurate descriptions or printing errors. We appreciate any feedback and will correct verified errors in future versions of the manual.
- At the time of publication, all descriptions, specifications, designs, and software programs contained in this document are valid. SunFlaser Optics Technology (Chengdu) Co., Ltd. reserves the right to modify the above information at any time without prior notice. If you have any questions regarding the use of this manual, please contact us promptly.

Safety

- The laser ranging module should not be viewed directly by the human eye. If observation is necessary, appropriate professional protective equipment must be used. Direct exposure to the laser beam is strictly prohibited to prevent potential eye injury.
- Avoid ranging on targets within the blind zone, especially highly reflective targets such as glass or polished metal surfaces. Avoid operating multiple ranging modules facing each other at close distances. Prevent high-energy laser sources from directly irradiating the receiving optics of the module. During assembly and debugging, keep the receiving lens covered at all times; otherwise, excessive echo signals may cause permanent damage to the detector.
- Pay attention to electrostatic discharge (ESD) protection: the electronic components of the laser ranging module are sensitive to electrostatic discharge. Do not touch any electronic components without proper protective measures.
- Do not touch the optical lenses directly with fingers or hard objects. Avoid using cleaning agents unless necessary, as they may cause irreversible damage to the lens coating and affect product performance.
- Do not use the product under direct sunlight. Do not store the product in highly polluted environments or outside the specified storage temperature range. Avoid exposing the laser ranging module to strong mechanical shocks, such as vibration, impact, or dropping. Avoid placing the module in environments with rapid temperature changes to prevent unpredictable effects on module performance.
- The laser ranging module adopts a certain level of sealing design. However, it is recommended to operate the product in environments with humidity below 80% and maintain a clean operating environment to extend the service life of the product.
- Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not attempt to tamper with or adjust the performance of the module.
- Operate the laser ranging module strictly according to the instructions provided in this datasheet. Avoid sending any operating commands that are not specified in this document.

Table of Contents

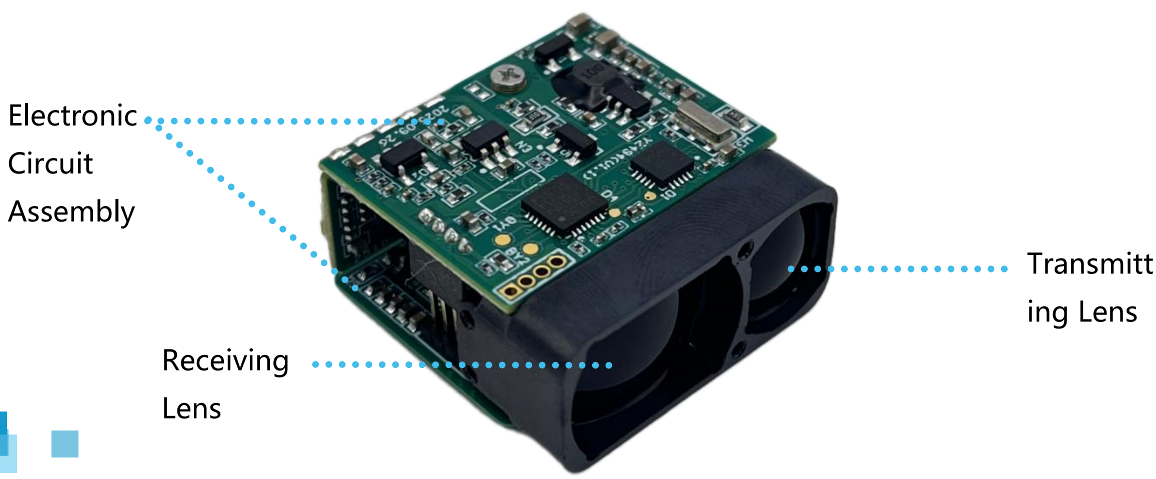
01 Product Overview

The SFA1000C Miniature Laser Ranging Module (hereinafter referred to as the laser ranging module) is designed to measure the distance to specific targets and provide accurate target distance information. It features high ranging accuracy and stable measurement performance under specified operating conditions and simple operation.The laser ranging module utilizes a 905 nm laser wavelength, and direct exposure of the laser beam to human eyes is strictly prohibited. The system communication is implemented through a TTL serial interface. With a modular design approach, the module achieves reduced size and weight while maintaining stable ranging performance.

02 Product Illustration

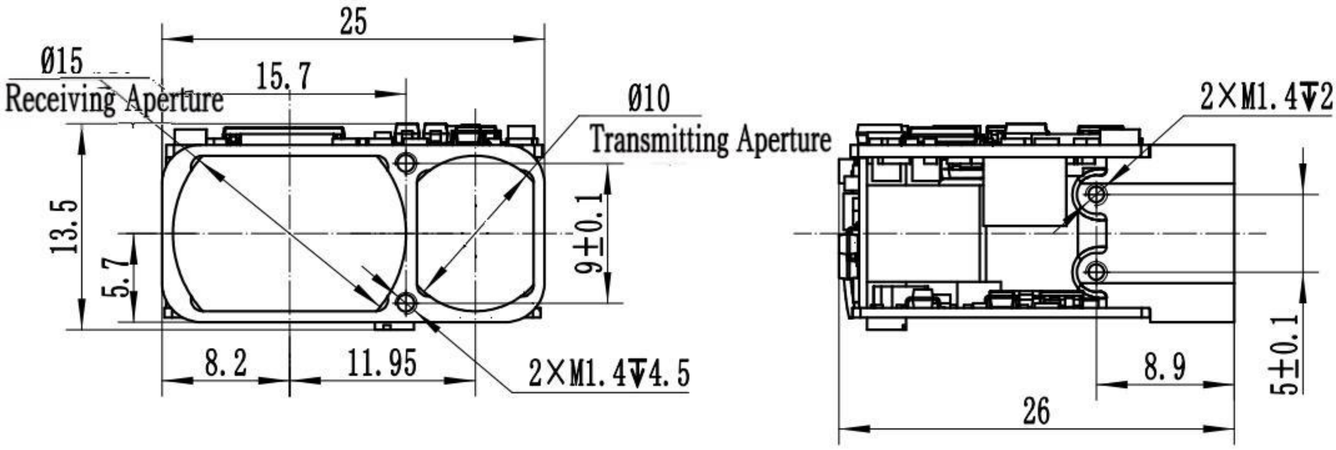
The product illustration of the laser ranging module is shown in the figure below. It mainly consists of the following components:

- Receiving Lens
- Transmitting Lens
- Electronic Circuit Assembly



03 Mechanical Dimensions

The mechanical and optical interface dimensions of the laser ranging module are shown in the figure below. Unit: mm.



1.Unit: mm;
2. Unless otherwise specified, tolerances shall comply with GB/T 1804-2000, grade m.

04 Product Performance

Item	Performance Specification
Item	Performance Specification
Operating Wavelength	905 nm ± 10 nm @ 25°C
Transmitting Aperture	Φ10 mm
Receiving Aperture	Φ15 mm
Ranging Capability	≥1000 m (Target: large target; visibility: ≥10 km; humidity: ≤60%; target reflectivity: 0.3)
	≥800 m (Target: forest; visibility: ≥10 km; humidity: ≤60%; target reflectivity: 0.3)
Blind Zone	≤5 m
Ranging Accuracy	≤±1 m (≤300 m), ≤±2 m (>300 m)
Measurement Accuracy Rate	≥99%
False Alarm Rate	≤2%
Ranging Frequency	Single Measurement
	Continuous Measurement (10 Hz for short-range measurement <300 m; 2 Hz for long-range measurement >300 m)
Weight	≤12 g
Dimensions	≤25 mm × 26 mm × 13.5 mm
Electrical Interface	Connector receptacle: 0.8WTB-6AWB-01 connector (Yueqing Huabao)
	Connector plug: 0.8WTB-6Y-2 connector
Supply Voltage	3.3 V – 5 V
Standby Power Consumption	≤0.8 W @ 3.3 V
Average Power Consumption	≤2 W @ 3.3 V @ 2 Hz
Operating Temperature	-20°C to +60°C
Storage Temperature	-30°C to +60°C
Environmental Compliance	Compliant with RoHS 2.0 requirements
Water and Dust Resistance	Leakage rate ≤1 sccm (air pressure: 30 kPa)
Function	Laser ranging function

Note: The nominal ranging distance and accuracy are achieved under conditions with visibility ≥10 km and relative humidity ≤60%. The ranging capability and accuracy of the module may vary depending on different environmental conditions and target characteristics.
Actual performance shall be subject to measured results. When the ranging module is used on a stabilized platform, ensure that the platform can maintain stable tracking of the target; otherwise, the ranging distance and accuracy of the module may be affected.

05 Electrical Interface

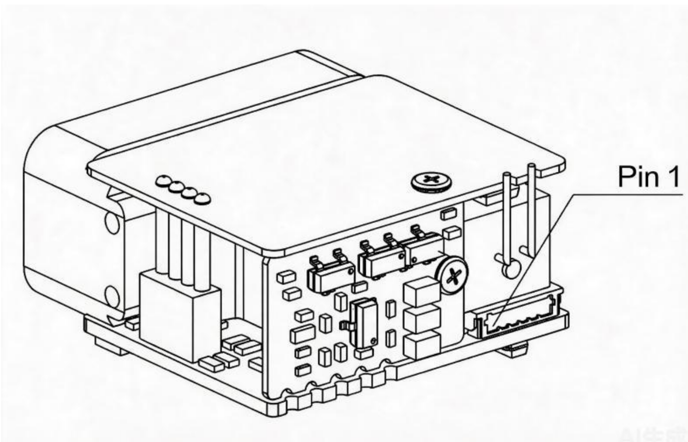
The electrical interface requirements are as follows:

- Supply Voltage: 3.3 V – 5 V;
- Standby Power Consumption: ≤0.8 W @ 3.3 V;
- Average Power Consumption: ≤2 W @ 3.3 V @ 2 Hz;
- The host system performs interconnection testing with the ranging module through the 0.8WTB-6Y-2 connector, which mates with the module-side 0.8WTB-6AWB-01 connector (Yueqing Huabao). The pin definitions of the power supply and communication ports on the ranging module side are shown in the table and figure below.

Pin Definition of Power Supply and Communication Ports on the Ranging Module Side

Pin No.	Label	Electrical Definition	Signal Direction
1	Power-EN		
2	TTL_RXD	Signal Input Port	Host to Ranging Module
3	TTL_TXD	Signal Output Port	Ranging Module to Host
4	NC		
5	VCC		
6	GND		

Electrical Interface Pin Diagram



06 Communication Interface

The communication interface requirements are as follows:

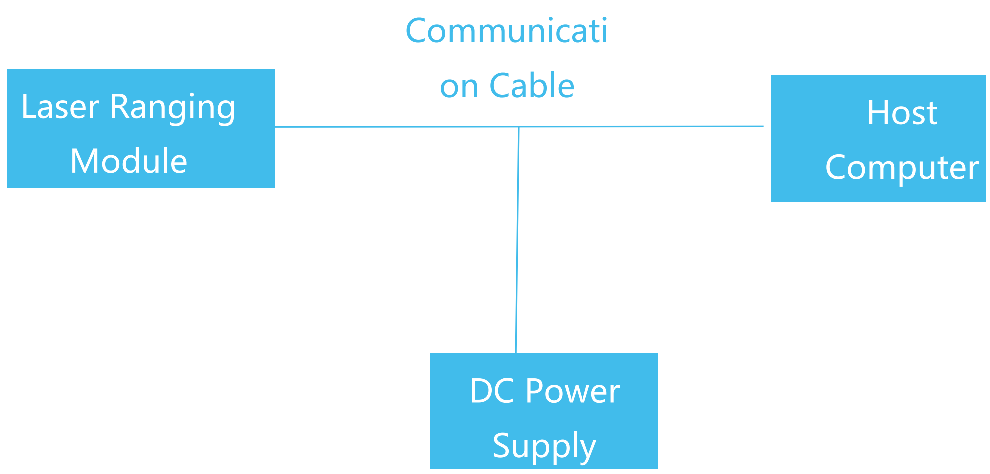
- Baud Rate: 115200 bps
- Transmission Format: Single-byte transmission format, including 1 start bit, 8 data bits, no parity bit, and 1 stop bit. The 8-bit data is transmitted with the least significant bit (LSB) first, followed by the most significant bit (MSB).
- Communication Protocol: Refer to Appendix 1.

07 Functions and Operation

1. Power-On Operation

● Before powering on, connect the laser ranging module, debugging cable, DC power supply, and host computer according to the electrical interface requirements as shown in the figure below (pay attention to the connector orientation).

● The module can be powered on by connecting the power supply.



Connection Diagram

2. Power-Off Operation

● Before powering off, ensure that all operating processes and tasks have been completed (after the single ranging result has been returned or continuous ranging has been stopped).

● After confirmation, disconnect the power supply to complete the normal shutdown process.

3. Function Description

Ranging Mode Operation:

● Send the "Single Ranging" command to the laser ranging module. The module performs a single distance measurement and reports the ranging status and distance value.

- Send the "Continuous Ranging" command to the laser ranging module. The module continuously performs distance measurements and reports the ranging status and distance values.
- Send the "Stop Ranging" command to stop the ranging operation.

08 Optical Window Selection

1. Material Recommendation

The recommended optical window material is H-K9L optical glass from Chengdu Guangming Optical Glass. H-K9L is one of the most commonly used colorless optical glasses, suitable for laser wavelengths ranging from 300 nm to 2100 nm. It features excellent physical properties and high cost-effectiveness.

2. Processing Recommendations

The wedge angle tolerance of the optical window should be minimized. A wedge angle tolerance of ≤3′ is recommended (tolerance grade ≤ Grade 7).
The optical surfaces of the window should be as smooth as possible. A recommended arithmetic average roughness (Ra) value is 0.012.

3. Coating Recommendations

For the 905 nm laser ranging module, it is recommended that the optical window be coated with an anti-reflection coating for the wavelength range of 855 nm to 955 nm, with a transmittance of ≥99.5%.Depending on the specific application environment, additional protective coatings such as hydrophobic coatings or hard coatings may be applied to the outer surface of the optical window. Other specifications shall comply with GJB2485-95, with a transmittance of ≥97%.

4. Optical Window Dimensions and Usage Recommendations

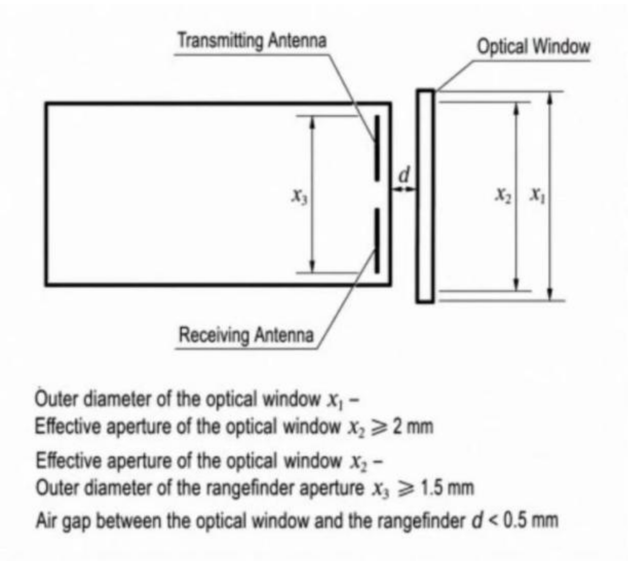
The effective aperture of the optical window depends on the specific product design. The window dimensions should ensure that the difference between the outer diameter of the optical window and its effective aperture is ≥2 mm, and the difference between the effective aperture and the outer diameter of the ranging module aperture is ≥1.5 mm. The schematic diagram is shown below.

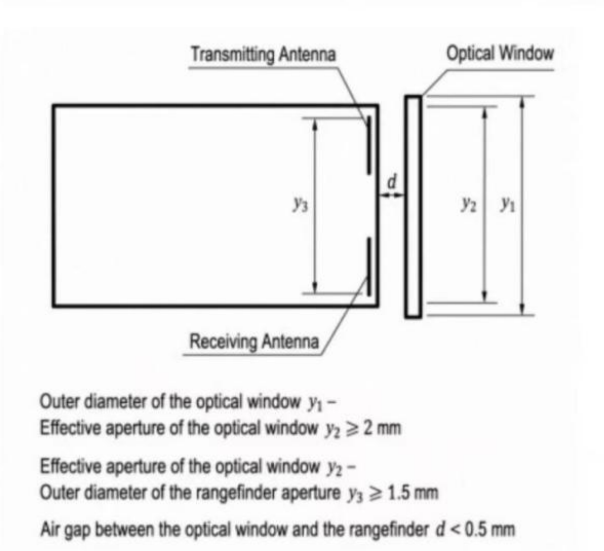
Since the optical window absorbs a certain amount of laser energy, the window thickness is recommended to be controlled within 2–4 mm depending on its dimensions.

Due to the high transmittance of the optical window, it is recommended that the transmitting optical axis be parallel to the normal axis of the window. Meanwhile, the air gap between the optical window and the ranging module should be less than 0.5 mm. The positional relationship between the optical window and the two lens barrels is shown in the figure below.

During use, the optical window must be kept clean and transparent. The optical surface should be free from stains, fingerprints, dust, and other contaminants.

Optical Window Dimensions and Placement Diagram





09 Maintenance Instructions

1. Packing List

The items supplied with the product are listed in the table below. (Optional accessories may not be included in the list. Please refer to the actual delivered items.)

No. No.	Item Item	Quantity Quantity	Remarks Remarks
1	Laser Ranging Module	1 unit	
2	Certificate of Conformity	1 copy	
3	Packaging Box	1 pcs	
4	Connector Plug	1 pcs	

2. Cleaning

2.1Cleaning of Optical Components

● Dust particles should be removed using an air blower.

● Fingerprints should first be removed by gently wiping with degreased cotton dipped in a small amount of alcohol-ether mixture, followed by cleaning with a clean lens cleaning cloth.

2.2Cleaning of Mechanical and Electronic Components

● Clean mechanical and electronic components with alcohol while the device is powered off, and allow them to air dry completely before use.

● Keep the ranging module, connector plug, and cables away from moisture and contaminants whenever possible.

● Ensure the equipment is completely dry before packaging.

3. Inspection and Maintenance

3.1 General Inspection

Visual inspection and power-on inspection shall be performed before the initial use of the product and after replacement of functional modules. For products in normal operation, only power-on inspection is required before use.

The visual inspection steps are as follows:

● Check whether the product appearance is normal.

● Check whether the cable connection is correct and ensure all connections are secure.

● The power-on inspection steps are as follows:

● Perform the power-on procedure and check whether the standby current is within the specified range.

● Start the self-test function to verify product operation.

● Perform the shutdown procedure after completing the inspection.

3.2 Periodic Maintenance

The laser ranging module does not require regular maintenance under normal operating conditions. Maintenance is required if the product has been stored in a dust-free environment for more than one year. Maintenance mainly includes general inspection and power-on inspection.

General inspection shall be performed with the product powered off. The procedures are as follows:

- All markings and numbers on the product and test cable connectors (plugs and sockets) shall be correct and clearly visible.
- All screws on the panel shall be properly tightened.
- Ensure that the optical glass is free from visible defects or contaminants that may affect normal observation, including spots, pits, water stains, mold, fingerprints, dust particles, cracks, or other attachments.

A complete inspection and maintenance procedure shall be performed with the laser ranging module powered on, including:

- Turn on the product power supply.
- Perform the power-on procedure and verify that the standby current is within the specified range.
- Start the product self-test function and complete the self-test procedure.
- Complete the shutdown procedure.

4. Fault Analysis and Troubleshooting

The laser ranging module is a precision product. In the event of a failure, the complete unit must be returned to the manufacturer for fault analysis, diagnosis, and repair. Self-repair is strictly prohibited. Common fault symptoms and troubleshooting methods are listed in the table below.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
No communication or abnormal communication	Power supply issue	Check whether the power supply has hardware or software wake-up enabled. Verify that the voltage setting is correct and ensure the	If the module operates normally after changing the voltage or power supply, the issue can be determined to	Set the correct voltage and current, or replace the power supply.

of the rangefinder module		protection current is properly configured. If the voltage is correct, try replacing the power supply and test again.	be caused by an incorrect power supply.	
	Connection cable issue	Check whether the wiring is secure, or replace the communication cable for testing (TTL communication cables should not be excessively long).	If communication is restored after rewiring or replacing the communication cable, the issue can be determined to be caused by a connection cable problem.	Rewire the connections or replace the communication cable.
		Check whether the wiring matches the interface pin definition.	If the module operates normally after confirming that the wiring matches the pin definition, the issue can be determined to be caused by incorrect wiring.	Reconnect the wiring according to the correct pin definition.
	Incorrect baud rate setting	1. Check whether the rangefinder status and baud rate settings are consistent with the communication protocol. 2. If the rangefinder supports baud rate configuration, change to the required baud rate and try communication again.	If communication operates normally after aligning the baud rate with the communication protocol, the issue can be determined to be caused by an incorrect baud rate setting.	Set the host computer baud rate to match the communication protocol.
	Incorrect communication command	Check the rangefinder status and verify whether the transmitted command is consistent with the communication protocol, and confirm that the command is in hexadecimal format.	If communication operates normally after confirming that the transmitted command matches the communication protocol, the issue can be determined to be caused by an incorrect communication command.	Send the correct command to the rangefinder module.
	RS422 driver not installed on the host computer.	Install the RS422 driver on the host computer.	If communication with the rangefinder module works normally after installation, the issue can be determined to be caused by the missing driver installation.	Install the driver.
	Main control program issue	If communication fails during integration with the main controller, first check whether standalone communication is functioning normally.	If standalone communication works normally, the issue can be considered likely to be caused by a problem in the main control program.	It is recommended to check whether there are issues in the system-level program (e.g., whether the parity/check bits are correct, and whether the host computer data parsing and

				display are implemented correctly).
	Other causes (e.g., circuit failure, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
The rangefinder returns invalid values.	Operational error / improper operation.	Check whether the receiving or transmitting lens is blocked, and ensure that there are no obstructions in the measurement environment (e.g., window glass, wall corners, etc.).	If the rangefinder returns normal readings after removing the obstruction, the issue can be considered likely caused by optical blockage.	Remove the obstruction.
		Whether the ranging axis is aligned with the optical sighting axis.	If the rangefinder returns normal measurements after realignment, the issue can be determined to be caused by misalignment between the ranging axis and the optical sighting axis.	Realign the ranging and optical axes and perform measurement again.
	Laser transmitter failure.	Check whether the current reading is normal. Observe whether the current fluctuates during ranging. Send a self-test command to check the returned data. Alternatively, use an infrared test card or a CCD camera to verify whether a laser spot is present.	If the emission self-test is abnormal and no laser spot is observed, the issue can be preliminarily determined to be caused by a laser transmitter failure.	Return to factory for repair.
	Other causes (e.g., circuit failure, detector, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
False ranging results.	Electrical noise interference.	Block or point the rangefinder receiving lens toward open space (or away from the sun) and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by optical noise interference. Further determine whether it is due to external ambient light or the system optical window (e.g., mismatch of wavelength band with the rangefinder, excessive tilt angle of the optical window, small aperture causing light blockage, or a contaminated optical window).		It is recommended to address power supply noise issues by adding a larger common-mode filter in series, replacing the grounding capacitor with a 10 nF capacitor, and increasing the detection threshold (note that this may reduce system performance).

	Communication noise interference.	Power the rangefinder module independently within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by electrical or communication noise interference. Power the rangefinder module independently within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by power supply noise interference.		It is recommended to address communication noise interference by adding a ferrite core (shielding magnetic ring) to the cable or increasing the threshold level (note that this may reduce system performance).
	Optical noise interference.	Operate the rangefinder module with communication only within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by communication noise interference. If false alarms still occur after the above steps, test the rangefinder module as a standalone unit (removed from the system and powered/communicated independently). If false alarms are still present, the issue can be determined to be caused by the rangefinder module itself or the power adapter.		It is recommended to address optical noise issues (optical window condition and test angle) and increase the detection threshold (note that this may reduce system performance).
	Other causes (e.g., circuit failure, etc.).	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
Insufficient performance capability of the rangefinder module.	Environmental conditions (weather-related factors).	Check whether the weather conditions include rain, snow, fog, or dust/sand, whether atmospheric visibility is too low, whether humidity is too high, or whether there are localized atmospheric turbulence effects, as well as any discrepancies between the environmental conditions and the test site.	If the ranging performance meets the technical specifications when the visibility is above the required threshold, the issue can be determined to be caused by environmental weather conditions.	Test the rangefinder performance under appropriate weather conditions.
	Optical window issue.	Determine, by checking the system configuration, whether the issue is caused by low optical window transmittance, incorrect wavelength band, contamination, small aperture causing light blocking, or excessive tilt angle. Remove the optical window and test the rangefinder module independently to observe whether its ranging performance is degraded. Also inspect whether the lens of the rangefinder is excessively contaminated.	If the ranging performance meets the specification requirements after removing the optical window, the issue can be determined to be caused by low transmittance of the host optical window. If the ranging performance meets the specification requirements after cleaning the rangefinder lens, the issue can be determined to be caused by low transmittance of the host optical window.	Replace the optical window according to the recommended selection and installation guidelines, or switch to a higher-performance rangefinder model. Clean the lens.

	Target size is too small or target reflectivity is too low.	Replace with another target. It is recommended to use targets such as buildings that are perpendicular to the optical axis, and avoid hollow or perforated targets as much as possible.	If the ranging performance meets the specification requirements after replacing with a higher-reflectivity or larger target, the issue can be determined to be caused by inappropriate target selection.	Replace the target, or switch to a higher-performance rangefinder model.
	Operating temperature exceeds the specified range of the rangefinder module.	When ranging performance degrades, send a self-test command to check the current operating temperature of the rangefinder module. After the temperature returns to normal, retest and observe whether the performance issue persists.	If the operating temperature exceeds the specified range when performance degrades, and the performance issue disappears after the temperature returns to normal, the issue can be determined to be caused by the operating environment exceeding the allowable temperature range of the rangefinder module.	Operate the device within the specified operating temperature range.
	Optical axis misalignment.	Check whether high/low temperature tests, thermal shock tests, shock tests, or vibration tests were performed prior to the degradation of ranging performance, or whether any drop or impact event occurred.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
	Other causes (e.g., circuit failure, low receiver sensitivity, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
Accuracy out of tolerance.	Large variation in target reflectivity.	Confirm whether the tested target is a high-reflectivity target, and determine whether the issue is large data fluctuation or accuracy out of tolerance.	If the tested target is highly reflective and the ranging accuracy returns to normal after replacing it with a suitable target, the issue can be determined to be caused by excessive target reflectivity variation.	Replace the target. The use of highly reflective test targets is not recommended.
	Test operation	Check whether the target has been properly aimed at.	If the ranging accuracy returns to normal after	Re-aim at the target.

	issue.		correctly aiming at the target, the issue can be determined to be caused by improper test operation.	
	Instrument issue.	Verify the accuracy of the calibration instrument (possible calibration error).	If the ranging accuracy returns to normal after replacing with an accurately calibrated instrument, the issue can be determined to be caused by the test instrument.	Replace the calibration instrument.
	Low optical window transmittance.	Check whether the ranging value is abnormally high.	If the ranging accuracy returns to normal after removing the optical window, the issue can be determined to be caused by the optical window.	Replace the optical window according to the recommended optical window selection and installation guidelines.

5. Packaging, Transportation and Storage Requirements

5.1Packaging

After unpacking, if the product needs to be stored again, it shall be repackaged using the original packaging. When the product needs to be returned to the manufacturer, the original packaging should be used whenever possible. If other packaging methods are adopted, they shall not cause any performance degradation or damage to the product.

5.2Transportation

Repackaged products may be transported by road, railway, air, or sea. During transportation, the packaged product shall be securely fixed to the transport vehicle to prevent impact, rough handling, rain, snow, or other adverse conditions.

10 After-Sales Service

If a product failure occurs and the complete unit needs to be returned to the manufacturer for fault analysis, diagnosis, and repair, SunFlaser provides a one-year warranty service and lifetime technical support from the date of product delivery. During the warranty period, SunFlaser will provide free replacement or repair for problems caused by product quality issues. For failures caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.
This product is a precision optical instrument. Please handle and protect the product carefully during use. If you have any questions regarding operation, maintenance, or service, please contact our after-sales support team at any time.

11 Contact Information

Website: www.sunflaser.com
Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)
Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1

Communication Protocol

1. Single Ranging Command

Note: Transmit checksum = Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7;

Receive checksum = Byte 1 + Byte 2 + Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7.

Command Sent to the Ranging Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	0xFF	0xFF	0xFF	0xFF	Checksum

Response from the Ranging Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	status	0xFF	DATA_H	DATA_L	Checksum

Status = 0: Single measurement failed;

DATA_H=0xFF，DATA_L=0xFF；

Status = 1: Single measurement successful;

DATA_H = High byte of the measurement result;

DATA_L = Low byte of the measurement result.

2. Continuous Ranging Command

Note: Transmit checksum = Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7;

Receive checksum = Byte 1 + Byte 2 + Byte 3 + Byte 4 + Byte 5 + Byte 6 + Byte 7.

Command Sent to the Ranging Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Freq	0xFF	0xFF	0xFF	0xFF	Checksum

Response from the Ranging Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Freq	status	0xFF	DATA_H	DATA_L	Checksum

Status = 0: Continuous measurement failed;
DATA_H=0xFF，DATA_L=0xFF；
Status = 1: Continuous measurement successful;
DATA_H = High byte of the measurement result;
DATA_L = Low byte of the measurement result.
Freq = 0x89: Continuous ranging mode;
Freq = 0xF9: Alignment calibration mode (the module returns the calibration status once after receiving the calibration command).

3. Stop Measurement Command

Command Sent to the Ranging Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	0xFF	0xFF	0xFF	0xFF	Checksum

Response from the Ranging Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	status	0xFF	0xFF	0xFF	Checksum

Status = 0: Failed to stop continuous measurement;
Status = 1: Continuous measurement stopped successfully.